

FLORA, IL, USA

WMO: 744658

Lat: 38.665N

Lon: 88.453W

Elev: 472

StdP: 14.45

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53889

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.9	(c) 11.9	(d) -2.4	(e) 4.9	(f) 8.5	(g) 2.3	(h) 6.3	(i) 12.7	(j) 26.6	(k) 41.4	(l) 24.3	(m) 41.3	(n) 6.0	(o) 300	(p) 0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	94.7	76.2	91.5	75.5	89.9	74.8	80.3	88.4	78.4	86.9	77.0	85.5	8.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	149.2	85.4	76.2	139.3	83.6	74.4	130.8	81.7	44.3	88.4	42.3	87.4	40.6	85.2	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.7	19.3	17.2		-0.5	97.2	6.8	3.5	-5.4	99.6	-9.3	101.7	-13.2	103.6	-18.1	106.1	
			DB	-1.1	81.7	6.3	2.4	-5.7	83.4	-9.4	84.9	-12.9	86.2	-17.5	88.0	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	31.5	34.1	45.7	56.4	66.1	74.9	77.1	75.5	69.2	57.5	45.5	36.3
	DBStd	18.41	11.49	11.40	10.84	9.04	8.30	5.65	5.48	5.30	6.77	9.09	9.47	9.67
	HDD50	1954	582	457	214	42	1	0	0	0	0	27	194	437
	HDD65	4665	1039	864	602	279	93	4	1	3	33	270	586	891
	CDD50	4119	8	13	82	233	499	747	841	790	576	260	59	11
	CDD65	1354	0	0	5	20	126	301	377	327	159	38	1	0
	CDH74	13288	0	0	39	214	1122	2912	3848	3141	1600	398	14	0
	CDH80	5250	0	0	4	25	347	1179	1679	1298	608	109	1	0
Wind	WSAvg	7.0	9.1	8.6	8.8	8.7	7.0	6.0	4.8	4.5	4.8	6.5	7.7	8.3
Precipitation	PrecAvg	44.5	3.5	2.4	3.5	4.2	5.2	4.2	3.9	3.5	3.1	3.4	4.2	3.0
	PrecMax	59.3	7.6	4.8	8.7	9.3	11.3	8.4	7.9	7.1	9.1	7.6	9.0	7.2
	PrecMin	35.7	0.4	0.4	0.9	1.3	1.4	1.1	1.8	1.3	0.2	0.8	0.2	1.5
	PrecStd	6.4	1.9	1.2	1.7	2.3	2.8	2.0	1.8	1.8	2.1	2.0	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.2	69.5	79.3	82.5	90.6	95.5	100.4	97.7	95.1	88.3	75.2	65.6
		MCWB	59.0	59.1	63.0	65.7	71.8	75.3	74.6	76.7	71.1	70.8	61.5	60.0
	2%	DB	59.1	63.1	72.6	79.7	86.8	92.5	94.6	93.4	90.7	82.4	70.1	60.1
		MCWB	54.4	56.9	61.0	63.8	71.1	75.0	78.3	75.8	73.0	67.0	60.6	56.4
	5%	DB	54.5	57.2	67.9	76.4	84.0	90.1	91.2	90.3	87.3	78.7	65.6	54.9
		MCWB	50.7	51.0	58.1	62.6	69.6	74.1	77.3	75.4	72.1	64.7	58.0	50.4
	10%	DB	48.4	51.8	63.5	72.4	81.1	86.9	89.2	87.4	82.5	73.4	61.3	51.0
		MCWB	44.7	46.8	56.2	61.0	68.1	72.6	75.9	74.6	68.7	63.4	54.6	47.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	62.4	65.3	69.6	75.8	81.1	83.9	83.7	77.7	73.8	66.9	61.8
		MCDB	63.5	67.4	74.3	78.6	85.3	89.2	91.8	93.9	87.3	82.6	72.7	64.6
	2%	WB	56.1	58.0	62.7	66.8	73.4	78.1	80.9	79.8	75.9	71.3	63.1	57.1
		MCDB	58.5	62.1	70.9	75.8	82.8	88.1	89.0	87.5	84.8	79.7	66.7	59.8
	5%	WB	51.2	52.2	60.0	64.6	71.6	76.3	79.1	77.8	73.9	68.4	59.6	52.0
		MCDB	53.4	56.1	66.1	72.9	80.4	85.6	87.3	86.2	82.0	74.8	65.2	54.4
	10%	WB	44.4	46.2	57.1	62.4	69.9	74.7	77.5	76.1	72.0	65.2	54.8	47.5
		MCDB	48.2	50.8	62.8	69.9	78.2	83.1	85.4	84.0	80.0	71.3	60.2	49.8
Mean Daily Temperature Range	5% DB	MDBR	15.2	15.9	18.2	20.5	19.2	19.8	19.4	20.6	22.8	21.7	18.3	14.2
		MCDBR	21.4	24.0	24.2	25.6	22.9	23.0	21.8	24.1	27.2	27.1	25.1	19.7
		MCWBR	17.9	18.6	14.9	14.0	10.0	8.8	8.5	9.6	10.5	12.9	16.6	16.2
	5% WB	MCDBR	19.5	22.7	21.4	21.8	20.0	19.2	18.9	20.0	21.6	21.5	21.2	17.8
		MCWBR	18.4	19.2	14.4	13.9	9.8	8.7	9.1	9.7	10.1	11.6	16.7	16.5
Clear-Sky Solar Irradiance	taub		0.309	0.323	0.360	0.402	0.438	0.439	0.464	0.465	0.412	0.360	0.320	0.307
	taud		2.453	2.400	2.364	2.247	2.204	2.224	2.157	2.141	2.281	2.425	2.484	2.471
	Ebn at Noon		275	286	285	278	269	268	259	256	264	269	269	267
	Edh at Noon		26	31	36	43	45	45	47	47	38	30	25	24
All-Sky Solar Radiation	RadAvg		615	865	1177	1552	1749	2027	1968	1781	1486	1071	720	523
	RadStd		61	92	109	90	151	132	124	101	104	111	80	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page